



LINAC Access Control Interlock System Validation and Verification Report

1.5.52.1 Rev.0

Date: 05-Aug-24

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Signature

Date

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REVISION HISTORY

<i>Revision</i>	<i>Date</i>	<i>Description</i>	<i>Author</i>
A	24-Aug-05	Issued For Review	Grant Cubbon
0	25-Aug-05	Issued for Use	Grant Cubbon

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1.0 PURPOSE

This report outlines the results from Validation and Verification (V&V) of the LINAC Access Control Interlock System (ACIS) conducted from August 15 – August 23, 2005.

2.0 BACKGROUND

The validation and verification of the Linac ACIS ensures that it is operating within the design specifications outlined in “1.2.37.2.Rev.0-Personnel Safety Lockup System-West [1] and in the 1.2.37.3.Rev.0-Personnel Safety Lockup System Micro84 Program-West [2]. The V&V procedure also verifies the “LINAC-LTB1 Lockup Procedure” 2.7.37.2 Rev.1B [3].

3.0 VERIFICATION AND VALIDATION RESULTS

3.1 SUMMARY

One non-conformance was found during the V&V testing that required immediate attention:

A problem with the deadbolt at Gate G10 occurred during lockup of zone 4.1. The gate could not be locked closed. The deadbolt was replaced by ETS staff and the V&V procedure was resumed.

Three non-conformances were reported (MKS 752-754) that involved burnt out Zone Lockup Light bulbs, defective Emergency Off Switches, and an open cover on the electrical box to a zone lockup light.

All non-conformances have been entered into the MKS tracking system. Each problem has been reviewed and repaired by the Controls and Instrumentation Development group. All checklists and the completed non-conformance forms are attached.

The Micro84 check was undertaken separately by two HSE staff on August 15 and August 23. No problems were encountered. Checklists for the Micro84 ACIS V&V are kept in a binder in room 1021.

4.0 REFERENCE MATERIALS

[1] Personnel Safety Lockup System-West 1.2.37.2.Rev.0

[2] Personnel Safety Lockup System Micro84 Program-West 1.2.37.3.Rev.0

[3] LINAC-LTB1 Lockup Procedure” 2.7.37.2 Rev.1B

5.0 ATTACHMENTS

LINAC Validation and Verification Checklists

Non-conformance reports (MKS 752-754)

EMERGENCY OFF AND PERSONNEL SAFETY SYSTEM

- One person in control room and one person in the room under test.
- The person in the room under test will press and reset each panic station checking for the following:
 - a) proper operation of station light.
 - b) proper operation of button locking device.
- The person in the control room will observe R1021.3-01, P1021.3-03 (P296) for the following:
 - a) proper operation of the horn.
 - b) proper operation of the indicating light.
- All faults must be recorded in maintenance log.

[illegible]

ROUTINE #1-1

PERSONNEL SAFETY INTERLOCK SYSTEM

Check each station and area lock-up light (see Figure #1 in the Personnel Inspection Lock-Up Procedure).

Zone #1, Area 0002

Station #1	✓	✓								
Lock-up indicator	✓	✓								

Zone #1, Areas 0001/0010/0011/00S2/10S2

Station #2	✓	✓								
Station #3	✓	✓								
Station #4	✓	✓								
Station #5	✓	✓								
Door (D2) Indicator	✓	✓								
Flashing Light	✓	✓								

Date:

23 MAY Aug 15/05

Initials:

PLU zol
EL U

Zone #2, Area 0008

Station #1	✓	—								
Lock-up Indicator	✓	—								

Zone#2, Areas 0009/00C1/0007

Station #2	✓	—								
Station #3	✓	—								
Station #4	✓	—								
Station #5	✓	—								
Station #6	✓	—								
Lock-up Indicator	✓	—								
Flashing Light	✓	—								

Date:

13 MAY 2005

Initials:

PK JC
E L

Zone#3, Areas 0003/01S6

Station #1	✓	✓								
Station #2	✓	✓								
Station #3	✓	✓								
Station #4	✓	✓								
Station #5	✓	✓								
Lock-up Indicator	✓	✓								

Zone#3, Areas 0014/0012/0013/0103

Station #6	✓	✓								
Station #7	✓	✓								
Station #8	✓	✓								
Station #9	✓	✓								
Station #10	✓	✓								
Station #11	✓	✓								
Station #12	✓	✓								
Lock-up Indicator	✓	✓								

Zone#3, Areas 0102/1007

Station #13	✓	✓								
Station #14	✓	✓								
Station #15	✓	✓								
Station #16	✓	✓								
Lock-up Indicator	✓	✓								
Flashing Light	✓	✓								

Date:

23 MAY Aug 15/5

Initials:

Rb
5 4

Zone#4, Areas 0004/0004.1

Station #1	✓	✓								
Station #2	✓	✓								
Station #3	✓	✓								
Station #4	✓	✓								
Lock-up Indicator	✓	✓								

Zone #4, Area 0006

Station #5	✓	✓								
Station #6	✓	✓								
Lock-up Indicator	✓	✓								

Zone #4, Area 0005

Station #7	✓	✓								
Lock-up Indicator	✓	✓								

Zone#4, Areas 00S1/01S1/10S1

Station #8	✓	✓								
Station #9	✓	✓								
Station #10	✓	✓								
Lock-up Indicator	✓	✓								
Flashing Light	✓	✓								

Date:

23 MAY 1990

Initials:

R. H.
R. H.

Zone #5, Areas 0108/1300

Station #1	✓	✓								
Station # <u>2,3,4,5</u>	✓	✓								
Lockup-Indicator	✓	✓								
Flashing Light	✓	✓								

Date:

13 MAY Ag. 1/10

Initials:

SL HL

CONTROL ROOM STATUS PANEL LIGHTS: Rack R1021.3-01 Panel P1021.3-09

ECS	✓	2								
LINAC Zone #1	✓	2								
South Access										
Storage Hut	✓	—								
South Access	✓	—								
Crane well Zone #2	✓	—								
EA2 Rear Storage	✓	—								
EA2 Rear Access	✓	—								
EA2	✓	—								
Switchyard	✓	—								
Power Supply Room	✓	—								
EA3	✓	—								
EA3 Access	✓	—								
EA3 Access Stairway										
Zone #4	✓	✓								
BLTB Tunnel										
Zone #5	✓	✓								
	✓									
All Zones	✓	✓								

Horns

Zones #1 Thru #5	✓	✓								
------------------	---	---	--	--	--	--	--	--	--	--

Date:

23 May Aug 15/05

Initials:

RLK JK
SR HL

ROUTINE #2

PERSONNEL SAFETY LOCK-UP

DPN2 104-02

1. Turn off two single pole breakers (Room 1021, P-1U, CCT #9,11).
Connect test lead to TB1-7 (test lead is 18" #14 wire with alligator clip on one end).
Reset aforementioned circuit breaker.

- See extra check sheet
2. Apply power to Flasher #1 at terminal #2. Flasher for zone #1 is now active.
Check lights for proper operation.
Repeat for Flasher #2 zone #2, Flasher #3 zone #3, Flasher #4 zone #4, and
Flasher #5 zone #5.

- 5
3. Remove the ~~4~~ wires at the bottom of TB1-25.
Apply power to each wire checking for proper operation of horns as listed below:

All
horns
working

Wire #25 - 1 - South Access, EA1, Linac Hall, EA3 Access
Wire #25 - 2 - EA3
Wire #25 - 3 - Power Supply Room
Wire #25 - 4 - Switchyard Room
Wire #25 - 5 - EA2

4. Turn off breakers from Step #1. Remove test lead; reconnect wires to TB1-25.
Reset breaker.

Date of Maintenance	Technician's Initials
May 24	42 2008
Aug 17/08	Grant Libban

22 2/24

- Zone 1 4 lights observed - all working
- Zone 2 4 " " - 3 working
- Outside bottom of crammell door, light not working
- Zone 3 9 lights observed - 8 working
- above E080003-02 not working
- Zone 4 5 lights observed - all working
- cover off junction box to horn in stairwell 1 - $\frac{1}{2}$ way between 5B + B
- Zone 5 6 lights observed - all working

ROUTINE #2-1

PERSONNEL SAFETY INTERLOCK SYSTEM

ZONE #1: Door or Gate Switches and Locks

Zone #1/Zone #3 Emg. Gate #1	✓	✓								
Area 0002/0001 West Wall Gate #2	✓	✓								
Area 0002/0001 North Wall Gate #3	✓	✓								
Zone #1/Zone #3 Radiation Door #1	✓	✓								
Zone #1/Zone #2 Door #1	✓	✓								
Zone #1/Zone #2 Emg. Exit Gate #4	✓	✓								
Area 0010/00S2 Door #2	✓	✓								
Area 10S2/1008 Door #4	✓	✓								
Area 10S2 Emg. Exit Door #3	✓	✓								

CS

✓

Date:

MAY 23/01/2015

Initials:

RL SC
SR

Zone #2: Door or Gate Switches and Locks

Zone #2/Zone #1 Emg. Exit Gate #4	✓	✓							
Zone #2/Zone #1 Door #1	✓	✓							
Area 0008/0009 Gate #6	✓	✓							
Zone #2/Zone #3 Gate #7	✓	✓							
Zone #2/Zone #4 Gate #8	✓	✓							
Area 00C1/0009 Door #5	✓	✓							
Crane Stair Access Door #12	✓	✓							

Zone #3: Door or Gate Switches and Locks

Zone #3/Zone #1 Emg. Exit Gate #1	✓	✓							
Zone #3/Zone #1 Radiation Door #1	✓	✓							
Zone #3/Zone #2 Gate #7	✓	✓							
Zone #3/Zone #2 Gate #11	✓	✓							
Area 0003/0013 Gate #15	✓	✓							
Zone #3/Zone #4 Door #6	✓	✓							
Zone #3/Zone #4 Gate #9	✓	✓							
Area 0012/0014 Door #7	✓	✓							
Area 0012/0013 Gate #13	✓	✓							
Area 0012/0013 Gate #14	✓	✓							
Area 0103/0102 Door #12	✓	✓							
Area 0102/01S6 Door #9	✓	✓							
Area 1006/1007 Door #10	✓	✓							

Control Room
cable duct
door

✓

Date:

May 23/01 Aug 17/05

Initials:

PLR
EHL

→ Zone 4 Green light in control room
did not turn on after lockup complete.

G10 doorbolt broke - repaired by NC
 08 ALTS 'touchy'
 - rubber off bottom of door

Zone #4: Door or Gate Switches and Locks

Zone #4/Zone #3 Gate #9	✓	✓								
Zone #4/Zone #3 Door #6	✓	✓								
Zone #4/Zone #5 Gate #16	✓	✓								
Zone #4/Zone #3 Gate #11	✓	✓								
Area 0004/0006 Gate #10	✓	✓								
Zone #4/Zone #2 Gate #8	✓	✓								
Area 0006/00S1 Door #8	✓	✓								
Area 00S1/0005 Door #11	✓	✓								
Area 01S1/0104 Door #17	✓	✓								
Area 10S1 Gate #17	✓	✓								

Elevator

Zone #5: Door or Gate Switches and Locks

Zone #5/Zone #4 Gate #16	✓	✓								
Area 0108/1300 Gate #18	✓	✓								

G21

G19

G20

Date:

May 23/01 Aug 17/05

Initials:

RI. 98
 SE 4

LINAC V&V EOS Checklist

EOS

	La Li	La Li	La Li	La Li
1	12	23	34	
2	13	24	35	
3	14	25	36	
4	15	26	37	
5	16	27	38	
6	17	28	39	
7	18	29	40	
8	19	30	41	
9	20	31	42	
10	21	32	43	
11	22	33	44	

No light on switch
(did not turn on)

Doesn't stay depressed until reset

Zone	Lockup Lights	POS Test ⁺ (Drops Lockup) Login Control Room
1	✓	✓
2	✓	✓
3	✓	✓
4	✓	✓
5	✓	✓

Zone 1 Gate key ban!
- needs to be replaced

LINAC V&V EOS Checklist

EOS

1	12	23	34
2	13	24	35
3	14	25	36
4	15	26	37
5	16	27	38
6	17	28	39
7	18	29	40
8	19	30	41
9	20	31	42
10	21	32	43
11	22	33	44



050423

6.16 Non-Conformance Report

Date: Aug 17 / 05

CLS Problem ID #: 753

Device or I/O Point: Zones 1-5 Flashing lights

Failure Description:	Not Working	Initials
Zone 2 - Light with EOS0009-04	→	JL
Zone 3 - Light with EOS0003-02	→	
Zone 3 - Light with EOS0003-04	→	
Cause:	Bulbs burnt out	NH
Resolution:	Replaced them	NH

Cc: HSE Manager
CLS Files

6.16 Non-Conformance ReportDate: Aug 17 / 05CLS Problem ID #: 754Device or I/O Point: LUS0151-02 (LUS9 Zone 4)

Failure Description:	Initials
The cover for the horn associated with LUS0151-02 is off. and the	SC
Cause: Screw has worked itself loose	NH
Resolution: Replace lid and tighten screw	NH

Cc: HSE Manager
CLS Files